AI-Driven Academic Risk & Dropout Early-Warning System

Mini Project - Review 2

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Abstract, Problem Statement & Objectives

Abstract

- Traditional systems find weak students only after exam results and keep data as static records.
- Digital Class Twin captures attendance, assessment, quiz and behavior data as events and keeps a live digital twin of each student.
- Risk is scored with explainable factors and shown on role-based dashboards for teachers, counselors and parents.

Problem Statement

- Weak students are identified only after examination or semester results.
- Static records do not show how performance changes over time.
- Attendance-threshold rules miss the complete academic situation.
- No mechanism for continuous monitoring and early intervention.

Objectives

1. Continuously collect attendance, assessment and behavior data
2. Maintain a Digital Twin per student: current state, history, sync status
3. Predict academic risk: Low, Medium, High
4. Predict future GPA using machine learning (planned)
5. Analyze trajectory: Declining, Stable, Recovering
6. Give explainable factors behind every prediction
7. Provide what-if simulation for interventions
8. Support early intervention by recording actions and outcomes

Literature Survey , Summary and Research Gap

20 papers reviewed (2021-2025): dropout prediction, early warning, educational data mining, explainable AI - listed in References

Research Gap

1. Many studies use static or institution-specific datasets.
2. Most focus mainly on prediction rather than continuous monitoring.
3. Limited integration of academic, attendance and behavioural events.
4. Few systems connect predictions directly with interventions.

Major Methods

1. Machine Learning: Random Forest, Decision Tree, SVM, Logistic Regression
2. Ensemble & Boosting methods for dropout prediction
3. Educational Data Mining and Learning Analytics
4. SHAP / Explainable AI for prediction interpretation
5. Digital Twin & graph-based student pathway modelling

Key Findings

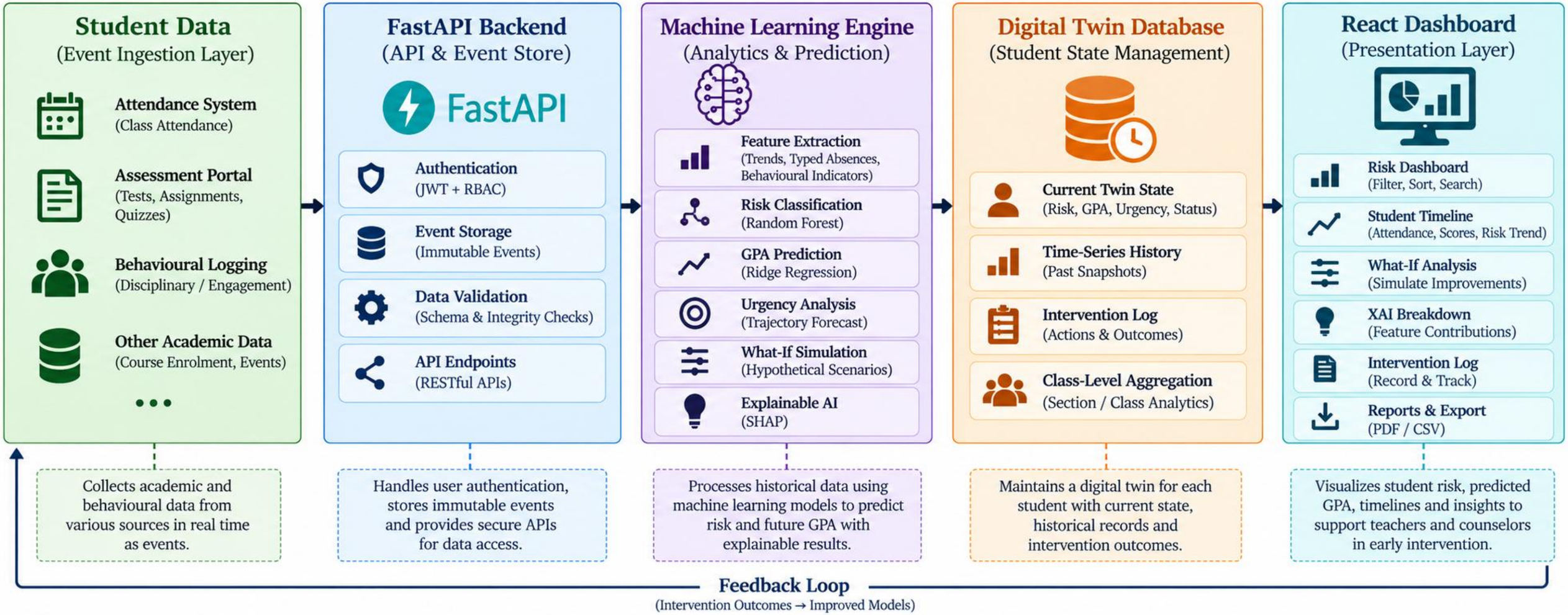
1. ML can predict academic performance and dropout risk at an early stage.
2. Attendance, academic performance, LMS activity and behaviour are important indicators.
3. Combining multiple student data sources improves risk prediction.
4. XAI helps identify the factors influencing predictions.

Relevance to Our Project

1. Our Digital Class Twin extends existing ML and Digital Twin approaches.
2. Uses event-driven continuous monitoring and time-series student state.
3. Provides Low/Medium/High risk, GPA prediction and trajectory-based urgency.
4. Uses SHAP, What-If simulation and intervention logging for actionable decisions.

Proposed System Architecture

Events → FastAPI ingestion → Rules & features → Risk engine → Twin & alerts → Database → REST API (JWT) → React dashboards



Design Methodology (Model, Algorithm, DB Design)

Algorithm

1. Receive typed event (attendance, marks, behavior)
2. Normalize, validate and store event
3. Recompute features: attendance %, marks trend, behavior
4. Compute composite risk score
5. Set twin state; save snapshot if state or risk changed
6. Raise alert on risk change, with explanation

Model

Composite risk score

score = 0.5 × Academic + 0.3 × Attendance + 0.2 × Behavior

≥ 75 Low → stable

50 - 74 Moderate → warning

< 50 High → critical

Rule/threshold-based, explainable model (risk_engine.py). Trained ML model is the next step.

DB Design

Users & access

User, UserSession, ParentChildMap, TeacherClassMap

Events & features

Event, StudentFeatures

Risk & twin

RiskScore, DigitalTwin, TwinSnapshot, Alert, InsightReview

SQLAlchemy ORM · SQLite (dev) · Alembic migrations

Implementation Details (Framework, Design)

Framework / Technology Stack

Layer	Technologies
Frontend	React, Vite, Axios, React Router, custom CSS
Backend	Python, FastAPI, Uvicorn, Pydantic
Database	SQLAlchemy ORM (SQLite dev), Alembic migrations
Security	JWT authentication, password hashing, role-based access
ML engine	Python; rule-based scoring (risk_engine.py, forecasting_engine.py)
Tools	Git/GitHub, Swagger (/docs), Postman

Project modules

- backend/ - FastAPI APIs, auth, event pipeline, DB models
- client-app/ - React + Vite role-based dashboards
- ml-engine/ - features, risk, forecasting, twin updater
- shared-contracts/ - event and twin JSON schemas

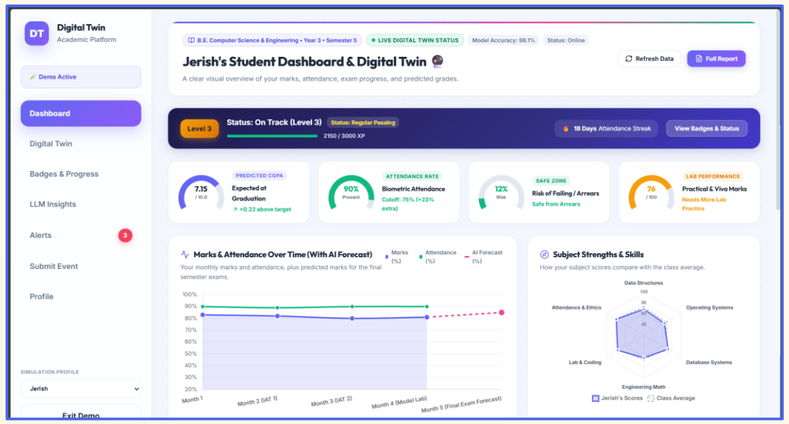
Role-based dashboards

- Student: Dashboard, Digital Twin, Insights, Alerts, Submit Event, Rewards, Profile
- Teacher: Dashboard, Class Overview, Risk Hub
- Parent: Dashboard, Child Twin, Academic Reports
- Admin: Dashboard, Audit Log, System Health

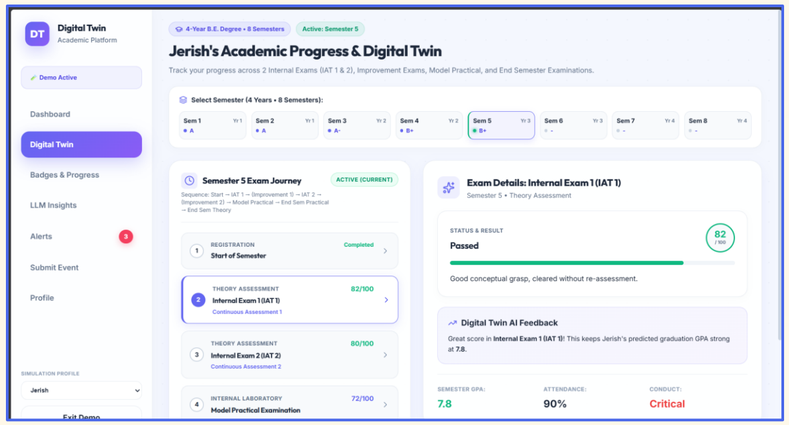
REST API routers

- auth, events, twins, insights, alerts, dashboard, profile, sessions, admin

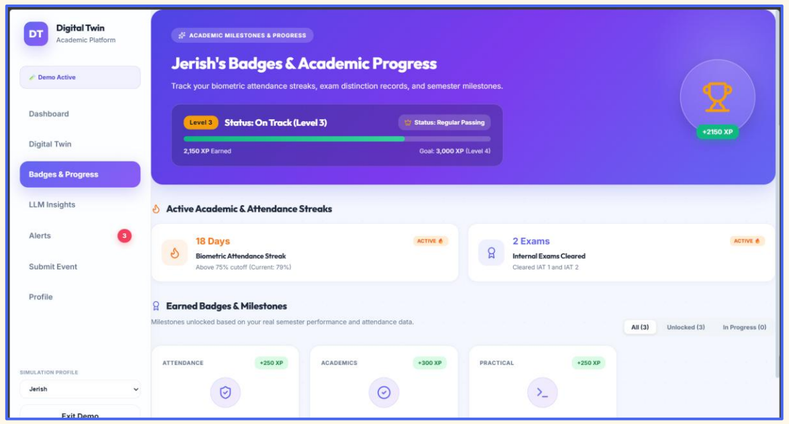
Implementation Module (Sample Output)



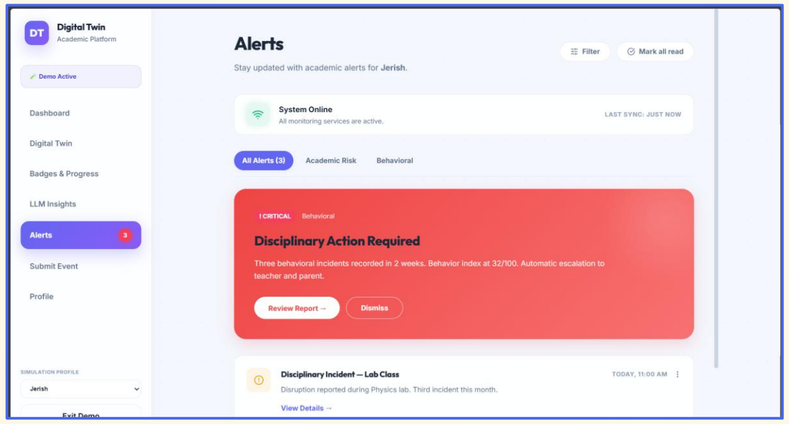
1. Student Dashboard



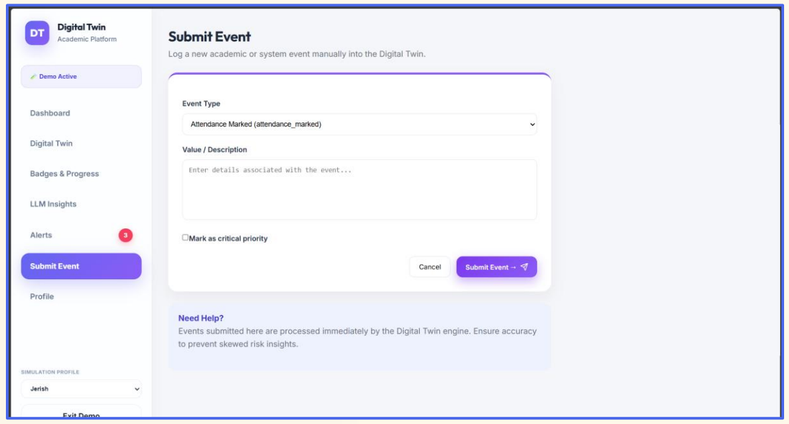
2. Digital Twin – semester progress



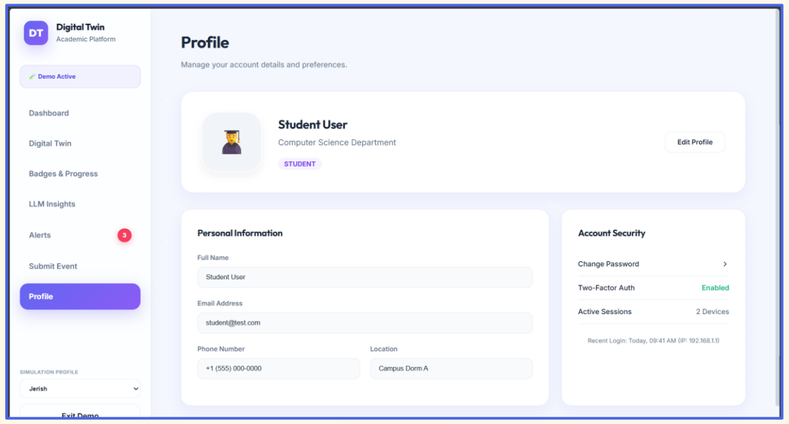
3. Badges & Progress



4. Alerts



5. Submit Event



6. Profile

Screenshots from the running prototype in demo / simulation mode.

Conclusion

Conclusion

- Working end-to-end prototype: events → features → risk → Digital Twin → alerts → dashboards
- Explainable risk scores with twin snapshots and trend history
- JWT login with role-based access for student, teacher, parent and admin
- Risk engine is currently rule-based

Future Work

- Trained ML model for risk prediction
- What-if simulator (shadow twin) for interventions
- Closed-loop calibration against real outcomes
- PostgreSQL database and deployment

References

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